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Seeing Surgical Skill: Vision-Based Hand Motion Analysis for Assessment in Orthopedic Trauma Training

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Master Thesis

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There's a fine line between fishing and just standing on the shore like an idiot.

— Steven Wright

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Generative AI Disclaimer

During the preparation of this thesis, ChatGPT and Gemini were used to assist with text formulation, coding support, and table formatting. All ideas, interpretations, analyses, and conclusions presented in this work are my own (and those of my collaborators where applicable).

Abstract

Objective and scalable assessment of surgical skill remains a central challenge in surgical education. While data-driven approaches have been widely studied in robotic and minimally invasive surgery, their application in open orthopedic and trauma procedures remains underexplored. This thesis investigates whether markerless hand tracking from video can be used to extract motion-based representations for objective skill assessment in this setting. Hand trajectories are extracted using a general-purpose tracking model and transformed into kinematic representations capturing both global motion characteristics and local temporal structure. These representations are evaluated using linear models, Bag-of-Words, and Multiple Instance Learning. The results show that markerless hand tracking provides motion data of sufficient quality for kinematic analysis. A large portion of the variance in expert-rated skill can be explained by a single global motion component capturing overall efficiency. Incorporating selected local motion descriptors yields a consistent improvement in predictive performance, reaching a Spearman correlation of 0.78 and achieving a level of consistency comparable to or exceeding individual expert raters. Overall, this work demonstrates the feasibility of vision-based, hand-centric motion analysis for surgical skill assessment and shows that predictive performance is primarily driven by informative motion representations, with global efficiency as the dominant factor and local structure providing complementary signal.

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Chapter 1

Introduction

The assessment of surgical skill is a fundamental component of surgical training and education. In modern training environments, opportunities for hands-on practice in real clinical settings are inherently limited, particularly for novice surgeons. Patient safety considerations and increasing procedural complexity restrict the extent to which trainees can gain experience through direct participation alone [10]. As a result, simulation-based training environments have become an essential component of surgical education, enabling repeated practice of technical skills under controlled conditions [10].

The effectiveness of such training, however, critically depends on the availability of timely and objective feedback that supports skill acquisition and guides improvement. Traditionally, surgical performance is evaluated through expert observation, often guided by structured rating systems such as the Global Rating Scale (GRS) [35] or procedure-specific checklists. While these methods provide clinically meaningful feedback, they are inherently subjective, time-intensive, and difficult to standardize across evaluators and institutions [28, 34]. As a result, there is increasing interest in developing objective, data-driven approaches that can quantify surgical performance in a reproducible and scalable manner [28].

A central challenge in this context is the acquisition of motion data that accurately reflects how a procedure is performed. In robotic and minimally invasive surgery, this problem is largely addressed through access to high-quality kinematic data recorded directly from surgical systems [1]. In contrast, orthopedic and trauma surgery are typically performed in open environments, where no standardized motion capture infrastructure exists. At the same time, these procedures involve specific interaction patterns, such as drilling, screwing, and manipulation of rigid structures, that differ substantially from those studied in minimally invasive surgical settings. This raises the question of whether meaningful motion-based assessment is feasible in such domains, and if so, how motion signals should be extracted and represented.

Existing work provides only limited answers to this problem. First, objective, data-driven skill assessment remains largely underexplored in orthopedic surgery, where most studies rely on expert-based video evaluation or isolated sensing modalities rather than end-to-end modeling approaches. Second, while recent advances in computer vision enable markerless hand tracking from video, only a very limited number of studies (e.g. [3, 18]) have explored its direct use for surgical skill assessment. Third, the few works that do leverage hand tracking typically rely on handcrafted proxies and statistical inference, without investigating predictive modeling or more expressive representations of motion. Taken together, this indicates a gap in the literature: despite the availability of general-purpose vision-based tracking methods, their potential for data-driven skill assessment in open surgical settings has not been systematically explored.

This thesis addresses this gap by investigating markerless hand tracking as a source of motion data for surgical skill assessment in orthopedic trauma surgery, using a dataset of video recorded simulated orthopedic procedures involving participants with varying levels of expertise. The core idea is to use a general-purpose hand tracking model (e.g., MediaPipe Hands [43]) to extract hand trajectories from video and to transform these trajectories into kinematic representations that can be used for quantitative analysis and predictive modeling of surgical skill. Such models operate efficiently on standard hardware in near real-time. This enables a fully vision-based pipeline that does not require specialized hardware and can be applied in realistic training environments.

Beyond signal extraction, the work further explores how surgical motion should be represented for skill modeling. While global descriptors such as path length or smoothness have been widely used in prior work, surgical performance unfolds over time and consists of heterogeneous motion patterns. This motivates the investigation of local motion representations that capture short temporal segments and their aggregation into higher-level descriptors. In particular, this thesis examines whether modeling skill as an emergent property of local motion structure provides additional information beyond global measures of efficiency.

The central thesis of this work is that surgical skill in orthopedic trauma procedures can be reliably characterized from markerless hand tracking data, and that this skill is primarily reflected as a global property of motion efficiency, with local motion patterns providing some complementary information. To evaluate this hypothesis, the study is structured around three research questions:

- **RQ1 — Signal Feasibility and Quality:** To what extent can markerless hand tracking provide motion data of sufficient stability and quality for kinematic analysis in orthopedic surgical tasks?
- **RQ2 — Global Motion Representation:** To what extent can global kinematic descriptors derived from hand motion explain variance in expert-rated surgical skill?
- **RQ3 — Added Value of Local Motion Structure:** Do representations of local motion structure capture skill-relevant information beyond what is already encoded in global motion efficiency?

The results show that markerless hand tracking provides sufficiently stable motion signals to support meaningful analysis in a controlled training setting. A large portion of the variance in expert-rated skill can be explained by a single global motion descriptor capturing overall efficiency. At the same time, incorporating selected representations of local motion structure yields a modest but consistent improvement in predictive performance beyond this global baseline. Together, these findings suggest that skill-related information is not confined to a small number of highly critical segments, but is distributed across the procedure, with local motion patterns providing complementary refinement of a dominant global efficiency signal. Notably, the predictive models achieve a level of consistency comparable to or exceeding individual expert raters relative to their aggregated mean score, while remaining bounded by the inherent variability of expert-based assessment.

The core contribution of this work lies in demonstrating the feasibility of vision-based, hand-centric motion analysis for orthopedic surgical skill assessment and in systematically evaluating how different levels of motion representation relate to predictive performance. By combining global kinematic descriptors, structured representations of local motion patterns, and learning-based aggregation methods within a unified framework, this thesis provides a comprehensive analysis of how surgical skill is encoded in hand motion and clarifies the relationship between global and local representations in this context.

Chapter 2

Related Work

Existing work can be understood along three key dimensions: how motion is captured, how it is represented, and how surgical skill is modeled from these representations.

2.1 Motion Capture in Surgery

A fundamental requirement for automated surgical skill assessment is the availability of motion data that accurately reflects the execution of a procedure. To reduce reliance on subjective expert evaluation, researchers increasingly employ data-driven methods to objectively measure surgical performance [28]. Existing approaches can broadly be categorized into sensor-based measurements and video-based methods.

Sensor-based approaches have traditionally dominated motion-based skill assessment due to their high precision and temporal consistency. In robotic surgery, kinematic data recorded directly from platforms such as the da Vinci system provide high-frequency, low-noise measurements and form the basis of widely used benchmarks such as JIGSAWS [1, 16].

In non-robotic settings, motion has been captured using external sensing technologies, including optical tracking systems and wearable sensors such as inertial measurement units, instrumented gloves, and force sensors. These approaches enable detailed measurement of hand kinematics and manipulation patterns [11, 34]. However, they require specialized hardware and controlled conditions, and introduce constraints related to sterility, ergonomics, and workflow, limiting their scalability in clinical environments [34, 41].

To address these limitations, vision-based approaches have gained increasing attention. These methods extract motion information directly from video, enabling markerless tracking without dedicated sensing hardware and allowing operation in realistic training and clinical environments [12]. However, they are subject to challenges such as occlusions, appearance variability, and view-

point dependence, which can affect signal consistency [9, 26, 40]. Within vision-based skill assessment, approaches can be distinguished based on the level at which motion is extracted, including instrument-centric, hand-centric, and direct video-based representations.

Instrument-centric approaches, which track surgical tools, are particularly prevalent in minimally invasive and robotic settings where instruments are consistently visible and closely linked to task execution. These methods provide structured and interpretable motion signals and are widely used for gesture recognition, phase detection, and skill assessment [9, 30]. However, they often rely on task-specific datasets and models, limiting their generalizability beyond controlled environments.

Hand-centric approaches instead aim to capture the surgeon's motion directly through hand or pose estimation, and are primarily explored in settings where the hands are clearly visible, such as open surgery or simulation [4, 7, 44]. A key advantage is the availability of general-purpose pose estimation models, such as MediaPipe, which enable efficient markerless tracking from monocular video [43]. More advanced models based on parametric representations, such as MANO, allow higher-fidelity reconstruction given sufficient input data [36].

Despite these advances, the use of vision-based hand tracking for surgical skill assessment remains limited. Only a small number of studies have explored this direction using general-purpose hand pose estimation frameworks [3, 18]. Related work has also explored vision-based hand tracking using color markers placed on the fingers to facilitate detection in controlled environments [19]. This suggests that hand-centric motion analysis remains an underexplored area despite the accessibility of modern tracking methods. At the same time, hand-based representations are inherently context-dependent, as not all observed motion is directly related to task execution. Movements may include auxiliary actions such as repositioning or interaction with the environment, requiring additional processing to isolate task-relevant signals [3, 4].

In addition to explicit tracking-based approaches, some methods operate directly on raw video, where motion is implicitly encoded in spatio-temporal image features. These approaches capture both movement and contextual information without requiring intermediate representations such as tools or hands [15, 40].

Overall, the choice of motion capture paradigm depends on the surgical context, data availability, and application requirements. The literature reflects a spectrum from structured, interpretable tracking-based representations to flexible end-to-end video models. Within this spectrum, hand-centric approaches offer a promising balance between scalability and interpretability, while introducing challenges related to variability and task relevance.

2.2 Motion Representation

Once motion has been captured, it must be transformed into a representation suitable for analysis. A motion representation refers to the structured transformation of raw observations, such

as trajectories, pose sequences, or video signals, into forms that can be compared and modeled. These signals are typically high-dimensional, temporally complex, and only indirectly related to performance. Raw motion data are therefore not used directly, but are transformed to impose structure, reduce redundancy, and emphasize relevant patterns [34]. This process may involve abstraction, dimensionality reduction, and temporal organization, and determines which aspects of motion are preserved for subsequent analysis [2].

Kinematic representations A widely used approach is to derive kinematic descriptors from trajectories or joint signals. Common features include path length, velocity, acceleration, jerk, and measures of smoothness or variability, which provide compact and interpretable summaries of motion [34]. Such descriptors have been shown to correlate with surgical expertise, with experts typically exhibiting more efficient, consistent, and smooth motion patterns [11, 13]. Their interpretability and low data requirements make them a standard baseline, particularly in robotic datasets with high-quality signals [11, 49].

Signal-based representations Motion can also be represented as a time series and analyzed in the frequency domain using transforms such as the discrete Fourier transform or discrete cosine transform [49]. These representations capture temporal structure, with low-frequency components reflecting global regularity and suppressing noise. Compared to higher-order derivatives, frequency-based measures are often more stable, and metrics such as entropy can quantify motion regularity [47, 49]. However, their effectiveness depends on signal quality and degrades with noisy tracking.

Early vision-based representations Early vision-based approaches relied on handcrafted spatio-temporal descriptors extracted directly from video. Methods such as space-time interest points detect salient motion regions and describe them using local image features [25]. These are typically aggregated using Bag-of-Words representations to form global descriptors [23, 46]. While fully markerless, such approaches operate on low-level visual patterns and do not explicitly encode kinematic structure.

Structured representations Motion can also be represented at different levels of temporal granularity by decomposing sequences into shorter segments. This enables modeling of temporal evolution rather than only global statistics. Segments can be organized into symbolic units such as gestures, motion primitives, or procedural phases [1]. These units are typically modeled as sequences, capturing temporal dependencies and repetition patterns. Approaches such as Hidden Markov Models encode transitions between states, while Bag-of-Words representations treat segments as elements of a vocabulary [8, 14]. While interpretable, these methods depend on reliable segmentation and annotation.

Learned representations In contrast to handcrafted descriptors, motion representations can be learned directly from data. Models operate either on kinematic time series or on raw video, where

image sequences are first transformed into feature embeddings capturing spatial information. These embeddings can be interpreted as multivariate time series and used to learn temporal patterns [39]. Such representations can be learned in supervised settings using skill labels, or in self-supervised settings such as autoencoders that model temporal evolution without explicit annotations [2]. Segment-based training is often used to increase the number of training samples [15]. While expressive, learned representations are typically less interpretable than handcrafted features and require sufficient data and appropriate supervision. In weakly supervised settings, treating all segments as equally informative may be problematic when motion signals are noisy or context-dependent.

Hand-centric representations While much of the literature focuses on tool-centric representations, the same principles apply to hand-centric motion extracted from video. Hand tracking provides sequences of joint positions that can be treated as kinematic time series and processed using similar feature extraction strategies. In practice, most work relies on handcrafted descriptors such as path length, velocity, smoothness, and coordination to characterize skill [5, 7, 29].

In vision-based settings, additional design choices arise. Some approaches focus on finer representations such as fingertip trajectories [19], while others use more stable landmarks such as the palm center to represent global motion [3]. These choices reflect a trade-off between sensitivity to fine motor behavior and robustness to noise.

More expressive representations, such as learned embeddings or end-to-end models operating on hand pose sequences remain largely unexplored. An approach has been investigated using IMU-based hand tracking data [31], but does not rely on vision-based inputs. As a result, existing work predominantly relies on handcrafted descriptors and statistical analysis, highlighting a gap in the literature.

More generally, the effectiveness of motion representations depends on data quality. Vision-based tracking is affected by noise, occlusions, and missing detections, which can distort derived features. Smoothing, filtering, and normalization are therefore commonly applied to ensure robustness [30].

Overall, motion representation determines how raw observations are transformed into structured signals that capture relevant aspects of performance. Existing approaches range from interpretable kinematic descriptors to learned embeddings, reflecting trade-offs in abstraction, data requirements, and robustness. The chosen representation directly influences subsequent modeling and skill assessment.

2.3 Skill Modeling from Motion Data

Given a motion representation, skill modeling aims to map motion-derived features to an estimate of surgical performance. This task is typically formulated as either classification, predicting discrete skill levels, or regression, estimating continuous scores. Such scores are commonly defined using

standardized assessment protocols such as OSATS [35] or GEARS [17].

Early approaches relied primarily on classical machine learning methods applied to handcrafted kinematic descriptors. Models such as support vector machines, random forests, and linear or logistic regression have been widely used due to their low data requirements and interpretability [11, 24]. Their performance, however, is largely determined by the quality and expressiveness of the input features, motivating the exploration of more advanced modeling approaches.

Many methods explicitly model temporal structure by operating on short motion segments represented as multivariate time series. This formulation is partly driven by limited dataset sizes, as segmenting procedures increases the number of training samples and enables analysis of local motion patterns [1, 24].

Segment-based approaches differ in how information is aggregated. In one formulation, segment features are aggregated accross time into a single representation before prediction using methods such as pooling, multiple instance learning or semantic aggregation [27]. In another formulation, predictions are made at the segment level and subsequently aggregated, for example through averaging or majorityvoting [2, 15].

Deep learning approaches typically operate on short temporal segments rather than full procedures. Common architectures include convolutional networks for spatial feature extraction combined with recurrent, temporal convolutional, or attention-based models to capture temporal dependencies [15, 27, 42]. For kinematic data, multivariate time series are processed using similar sequence modeling techniques [39, 42].

However, these approaches are constrained by the nature of available supervision. In many datasets, only global skill scores are provided, without segment-level annotations [1, 24]. A common strategy is to assign global labels to all segments, implicitly assuming that each segment is equally informative [2, 15]. This assumption may be problematic in settings with heterogeneous motion or noisy signals, where only subsets of segments are relevant for skill assessment [48].

In the specific case of hand-centric motion extracted from video, the range of explored modeling approaches remains limited. Most work relies on handcrafted kinematic descriptors combined with classical models, reflecting both limited dataset sizes and the variability of vision-based tracking [4, 18, 19]. Learning-based approaches operating directly on hand motion are comparatively rare, particularly for vision-based data. For example, Nguyen et al. [31] use a CNN–LSTM model on sensor-based hand motion data, but similar approaches remain largely unexplored for markerless hand tracking.

Overall, skill modeling methods range from interpretable classical models to sequence-based deep learning approaches that explicitly capture temporal structure. Their effectiveness depends on the underlying motion representation and the nature of available supervision, particularly the distinction between global labels and segment-level information. These considerations motivate

approaches that combine local motion analysis with flexible aggregation mechanisms, as explored in this work.

2.4 Applications in Orthopedic and Trauma Surgery

Compared to robotic and minimally invasive surgery, where standardized datasets and motion signals are widely available, automated skill assessment in orthopedic and trauma surgery remains relatively underexplored. While motion- and video-based approaches have been studied in other open surgical domains, their adoption in orthopedic contexts is limited and lacks standardized benchmarks.

Orthopedic procedures also exhibit distinct interaction patterns, such as drilling, screwing, and manipulation of rigid structures. These tasks involve different motion dynamics and coordination strategies compared to those studied in laparoscopic or robotic surgery. As a result, methods developed in other domains may not transfer directly, motivating the need for domain-specific approaches.

Existing work in orthopedic settings has largely focused on subjective or semi-structured evaluation, including expert-based video assessment and feasibility studies [20]. Some studies have explored technical components such as instrument detection or sensor-based motion capture [32, 33]. Recent work has also investigated tool-tracking-based approaches for performance assessment in orthopedic training environments [37], highlighting the potential of motion-based analysis in this domain. However, such approaches typically rely on instrumented setups and controlled conditions, which may limit their scalability and applicability in realistic clinical or training settings.

Overall, there is a lack of approaches that integrate motion extraction with objective, data-driven modeling of surgical skill in orthopedic and trauma surgery using minimally intrusive sensing modalities. This gap highlights an opportunity for developing scalable, vision-based methods that can operate in realistic training environments without requiring specialized hardware.

Chapter 3

Dataset and Experimental Setup

The data used in this work originates from a previously conducted experimental study on surgical skill assessment in a simulated orthopedic trauma training environment [38]. The original study employed optical tool tracking, whereas the present work focuses on a vision-based hand tracking approach. The setup was designed to replicate essential aspects of orthopedic trauma procedures using synthetic bone models and standard surgical instruments. An overview of the training setup is shown in Figure 3.1.

(a) (b)

Figure 3.1: **a)** Setup with synthetic bones secured in a custom holder and **b)** tools used to perform the procedure equipped with optical markers. Source: Stauffer et al. [38].

Participants performed three standardized fracture fixation tasks on synthetic bone models, corresponding to a simple oblique radial shaft fracture, a simple fibula fracture, and a multifragmentary ulna shaft fracture. Each task was treated using appropriate plating techniques (compression, neutralization, and bridge plating, respectively).

All procedures were recorded on video at a resolution of 1920×1080 pixels and a frame rate of 30 frames per second. In total, 83 procedures performed by 28 participants were included in the dataset

(28 radius, 28 fibula, and 27 ulna procedures). The cohort comprised individuals with a wide range of surgical experience, including medical students, residents, and senior physicians. The mean surgical experience of the participants was 6.6 years (SD = 8.6), resulting in a broad distribution of skill levels suitable for learning-based assessment.

Surgical performance was evaluated independently by four expert trauma surgeons (three senior consultants and one attending physician). All recordings were anonymized and assessed in a blinded manner. Skill was rated using the Global Rating Scale (GRS), which evaluates multiple aspects of surgical performance, including instrument handling, motion efficiency, and procedural flow [35]. The soft tissue handling domain was omitted as it was not applicable in the simulated setting.

Across all procedures, a total of 332 individual ratings were obtained. In this work, the target variable is defined as the mean of the four expert ratings per procedure. The resulting aggregated GRS score had an average of 49.9 (SD = 9.36), with values ranging from 33.75 to 67.75 points. Case-specific averages were 47.5 (SD = 9.17) for the radius task, 50.0 (SD = 10.2) for the fibula task, and 52.4 (SD = 8.29) for the ulna task, indicating moderate differences in scoring between procedures.

Inter-rater agreement at the individual rater level was moderately high. The reliability of a single expert rating, quantified using the intra-class correlation coefficient (ICC(2,1)), was 0.71. Consistent with this, individual ratings differed from the aggregated mean on average by 4.8 GRS points (SD = 3.5), indicating non-negligible variability between raters.

Averaging across raters substantially improved the reliability of the resulting score. The aggregated rating achieved an intra-class correlation of ICC(2,k) = 0.91 (k=4), reflecting strong consistency of the mean GRS score used as the prediction target in this work. The corresponding standard error of measurement (SEM) was approximately 2.9 GRS points, indicating the remaining uncertainty associated with the aggregated expert assessment. This value provides an estimate of the inherent noise level in the target signal and therefore a practical lower bound on the achievable prediction error for data-driven models.

This dataset presents several characteristics relevant for subsequent modeling. The diversity in participant experience results in a broad distribution of skill levels, while differences between procedures introduce task-specific variation in motion patterns. At the same time, the observed inter-rater variability highlights the inherent uncertainty of expert-based skill assessment, which constrains the achievable performance of predictive models.

Chapter 4

Methodology

4.1 Tracking Pipeline

The extraction of hand motion trajectories consisted of four main stages: (1) video preprocessing and hand landmark extraction using MediaPipe, (2) selection of stable palm landmarks to represent global hand motion, (3) trajectory post-processing to correct label inconsistencies and fill short detection gaps, and (4) temporal smoothing to reduce tracking noise.

Video Preprocessing and Landmark Extraction All surgical simulation recordings were manually trimmed to align with the start and end of the surgical task. This ensured that only relevant task execution was retained while excluding setup and cleanup phases. No additional manual trimming was performed within the videos in order to preserve realistic variability and noise.

Hand landmarks were extracted using the MediaPipe Hands framework [43] (version 0.10.21). The framework combines a palm detection model with a landmark regression model that estimates 21 hand landmarks per frame. In addition to landmark coordinates, a handedness classification score is provided for each detected hand. To improve efficiency, the detected cropped region from the previous frame is reused for landmark estimation, while the palm detector is only invoked when tracking fails. Although MediaPipe provides three-dimensional coordinates, only the two-dimensional image-plane coordinates were used. The depth component represents a relative approximation and is more sensitive to noise. Given the fixed camera setup, two-dimensional trajectories provide a sufficiently stable representation for kinematic analysis. The original frame rate of 30 fps was retained to preserve temporal resolution.

Palm Landmark Selection The MediaPipe model outputs 21 anatomical landmarks describing the wrist, palm, and finger joints. For kinematic analysis, six palm landmarks (0, 1, 5, 9, 13, and 17) were retained, as shown in Figure 4.1(a). These landmarks form a relatively rigid structure that approxi-

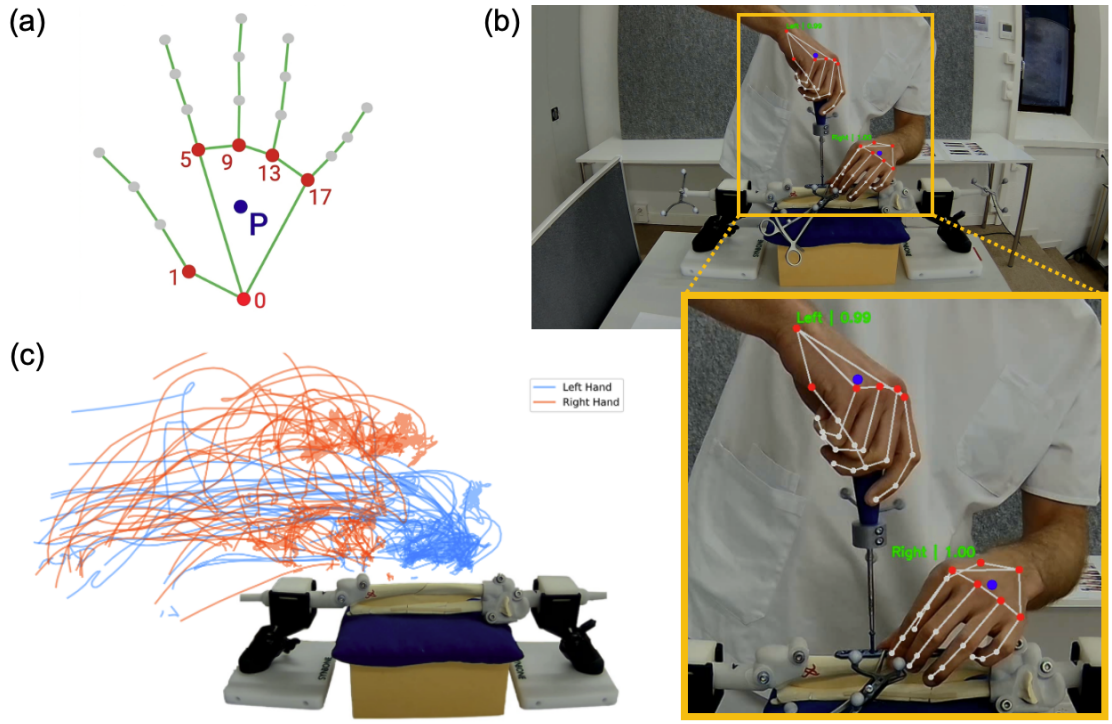

Figure 4.1: Hand trajectory extraction pipeline. **a)** MediaPipe hand model keypints, highlighting the palm landmarks used to calculate the palm center P. **b)** Example frame with detected hand landmarks. **c)** Processed palm-center trajectories for left (blue) and right (orange) hands.

mates global hand motion while avoiding highly articulated finger joints. Finger landmarks are more susceptible to noise and occlusion, whereas palm landmarks provide a more stable representation of hand movement. This is particularly relevant in orthopedic tasks, where motion is dominated by tool manipulation rather than fine finger articulation. A single palm center was computed as the mean position of the selected landmarks:

$$c_x = \frac{1}{N} \sum_{i \in P} x_i, \qquad c_y = \frac{1}{N} \sum_{i \in P} y_i,$$

where P denotes the selected palm landmarks.

A similar representation has been reported by Atazadah et al. [3]. Additionally, landmarks 0, 5, and 17 were retained to define a triangular configuration for approximating hand orientation.

Trajectory Post-Processing The raw trajectories were processed to improve temporal consistency and reduce noise. This included correction of label inconsistencies, handling of short detection gaps, and temporal smoothing.

MediaPipe assumes a horizontally flipped input for handedness prediction, which is inconsistent

with the camera perspective used in this setup. Detected left and right labels were therefore swapped to match the recording configuration. A temporal consistency filter was then applied to ensure that each frame contained at most one left and one right hand detection. When multiple detections were present, the most plausible candidate was selected based on the MediaPipe handedness score and spatial continuity relative to the previously observed hand position.

Short tracking gaps of up to 0.1 s were filled using linear interpolation, while longer gaps were treated as missing data to avoid introducing unrealistic trajectories. Based on this threshold, trajectories were segmented into continuous sequences.

Finally, a two-stage filtering approach was applied to smooth the trajectories. A median filter (kernel size 3) was used to remove impulsive outliers, followed by a Savitzky–Golay filter with a window of 0.3 s and a second-order polynomial. This window balances noise reduction and preservation of motion dynamics, ensuring that meaningful movement patterns are retained [9].

The resulting smoothed palm center and landmark trajectories form the basis for subsequent kinematic feature extraction.

4.2 Global Motion Framework

4.2.1 Global Kinematic Feature Extraction

Global motion descriptors were extracted from the smoothed hand trajectories. For each hand, the palm center trajectory was used as a stable representation of overall hand position. From this trajectory, frame-to-frame displacement was computed while respecting trajectory segmentation to avoid discontinuities caused by tracking gaps. Temporal derivatives, including velocity, acceleration, and jerk, were subsequently derived based on the video frame rate, following common practice in motion-based surgical skill assessment [11, 34].

With these kinematic signals, a set of global motion descriptors was computed to capture different aspects of surgical motor performance. The features were organized into five conceptual domains: translational motion, trajectory smoothness, directional dynamics, hand orientation dynamics, and bimanual coordination (Table 4.1). Translational motion descriptors capture the extent and intensity of hand movement through measures of displacement, velocity, and acceleration. Smoothness metrics characterize the fluidity and control of the trajectory. Common measures include jerk-based smoothness and movement segmentation [4, 11]. Directional dynamics metrics capture changes in movement direction, reflecting the degree of path irregularity and corrective adjustments. Finally, bimanual coordination metrics describe the temporal and spatial interaction between both hands.

Rotational and pose-related descriptors were derived from a rigid palm representation defined by the triangle formed by landmarks 0 (wrist), 5 (index finger base), and 17 (pinky base) (Figure 4.1(a)). This triangular structure approximates the orientation of the palm. This structure provides a stable approximation of palm orientation in the image plane, enabling the computation of orientation

entropy, angular displacement, and pose variability measures. A complete overview of the extracted motion descriptors is provided in Table 4.1, and the mathematical definitions are provided in Appendix 8.1.2.

4.2.2 Global Feature Selection and Baseline Modeling

To identify a robust global skill signature and construct a baseline predictive model, a hierarchical statistical pipeline was applied. Pearson correlation coefficients were computed between each extracted feature and the ground-truth GRS scores, providing an initial assessment of which motion descriptors exhibited the strongest association with surgical skill. Many of the highest-ranking features showed strong intercorrelations, reflecting overlapping aspects of movement efficiency. To reduce redundancy and construct a compact global representation, Principal Component Analysis (PCA) was applied to this subset of highly correlated features. The first principal component (PC1) was retained as a global efficiency index representing the dominant variation pattern in motion efficiency.

Separate PCA models were computed using (i) dominant-hand features only and (ii) combined left and right hand features. The configuration yielding the highest explained variance for PC1 was selected as the global representation. This comparison was motivated by the right-hand dominance of participants and the predominance of instrument manipulation in the dominant hand.

A baseline regression model was then constructed using the global efficiency index (PC1) as the primary predictor of GRS scores. To account for structural differences between surgical tasks, case type was included as a categorical variable representing the three simulated procedures. Model estimation used ridge regression with cross-validated regularization (RidgeCV) to stabilize coefficient estimates and mitigate potential overfitting given the limited dataset size. The resulting baseline model can be expressed as:

$$GRS \sim PC1 + CaseType,$$

where CaseType is encoded as a one-hot categorical variable representing the three procedure types.

To assess whether nonlinear relationships between motion features and skill scores could provide additional predictive power, a small multilayer perceptron (MLP) model was additionally evaluated using the same input variables. The network consisted of a lightweight fully connected architecture with one hidden layer of 16 units using a standard nonlinear activation function (LeakyReLU). To reduce overfitting, dropout, L2 weight decay, and early stopping based on training loss stagnation were applied. In addition, an ensemble of five independently initialized models was trained and their predictions averaged, with weight averaging applied over the final training epochs to further stabilize the model.

To investigate whether additional motion descriptors contained information beyond global motion efficiency, a residual analysis was performed. First, the baseline ridge model using the global efficiency index (PC1) and case type was fitted within each leave-one-subject-out (LOSO) training

Table 4.1: Overview of global motion descriptors extracted from hand trajectories. Features are grouped by conceptual domain and computed from the smoothed palm center trajectory or the palm triangle representation (landmarks 0, 5, 17).

Domain	Feature	Description
Translational Motion		
	Total Path Length	Sum of frame-to-frame displacement of the palm center
	Mean Velocity	Average translational velocity along the trajectory
	RMS Acceleration	Root-mean-square acceleration magnitude
	Path Rate	Total path length divided by task duration
Trajectory Smoothness		
	LDLJ Smoothness	Log dimensionless jerk measuring movement smoothness
	Number of Movement Units (NMU)	Number of prominent peaks in the velocity profile
Directional Dynamics		
	Mean Absolute Angle Change	Average directional change between consecutive displacement vectors
	Angular Direction Switches	Number of sign reversals in trajectory curvature
Hand Orientation Dynamics		
	Total Angular Path	Total accumulated change in palm orientation
	Orientation Entropy	Entropy of the palm orientation distribution
	Pose Variability	Variability of the palm triangle area (pose stability)
	Rotation over Distance	Angular displacement normalized by translational path length
Bimanual Coordination		
	Velocity Correlation	Correlation between left and right hand velocities
	Movement Overlap Ratio	Fraction of frames where both hands move simultaneously
	Inter-hand Distance Stability	Root-mean-square change of distance between hands
	Velocity Ratio	Ratio of mean velocities between left and right hands

fold. The residuals of this model were then computed and used as the target for further analysis. Each remaining motion descriptor was evaluated for its ability to explain residual variance by fitting a ridge regression model including the baseline predictors and the candidate feature. The additional explanatory power of each feature was quantified using partial R^2:

$$R^2_{\text{partial}} = \frac{\mathscr{L}_{\text{base}} - \mathscr{L}_{\text{aug}}}{\mathscr{L}_{\text{base}}}.$$

This quantity represents the proportional reduction in residual variance achieved by adding the candidate feature to the baseline model.

To assess whether observed improvements could arise by chance, an empirical residual shuffling test was performed. For each fold, the baseline model was first used to generate residuals for the entire dataset. These residuals were then randomly permuted to construct a null target with the same variance structure but no meaningful relationship to the candidate features. The same regression procedure was repeated using this shuffled target, producing a distribution of partial R^2 values under the null hypothesis. Comparing the observed partial R^2 values against this null distribution allowed potential secondary skill signals to be distinguished from improvements expected due to random correlations.

4.2.3 Evaluation Protocol

Model performance was evaluated using leave-one-subject-out cross-validation (LOSO-CV). In each fold, all recordings belonging to one surgeon were held out for testing, while models were trained on the remaining surgeons. This evaluation scheme prevents information leakage between training and testing sets and ensures that model performance reflects the ability to generalize to previously unseen surgeons. Importantly, the entire modeling pipeline was fitted independently within each cross-validation fold. All preprocessing steps, including feature standardization, dimensionality reduction using Principal Component Analysis (PCA), and unsupervised procedures such as k-means clustering for the motion vocabulary construction, were estimated exclusively on the training data of the current fold. The learned transformations were then applied to the held-out surgeon data without refitting.

All models evaluated in this study, including the global baseline model, residual feature analysis, and subsequent local movement models, were assessed using this protocol. Performance metrics were computed by aggregating predictions across all cross-validation folds and comparing them to the corresponding mean expert rater GRS scores.

4.3 Local Motion Framework

A key design choice in this work concerns how local motion information is related to global skill assessment. Rather than predicting skill at the level of short temporal segments, local motion is

first represented through window-level feature extraction and subsequently aggregated to obtain an instance-level representation for prediction. This design reflects the fact that only a single global skill score is available per procedure and cannot be reliably assigned to individual motion segments. In contrast, prior work often performs prediction at the segment level by assigning the same skill label to all temporal windows within a trial [2, 15], implicitly assuming that skill-related information is uniformly distributed over time. In the present setting, however, motion signals derived from vision-based hand tracking are more variable and may include segments that are only weakly related to task execution. Aggregating local representations rather than local predictions therefore provides a more robust formulation, allowing skill to be inferred from the distribution of motion patterns while reducing the influence of noisy or non-informative segments. This formulation also enables the use of aggregation strategies that assign different levels of importance to individual motion segments, as explored in the following sections.

4.3.1 Local Kinematic Feature Extraction

Hand trajectories were analyzed using a sliding window approach to capture short segments of motion corresponding to elementary surgical hand movements. Fixed-length windows of 1.5 seconds were extracted with a step size of 0.5 seconds, resulting in overlapping windows that provide dense temporal coverage while preserving local motion structure. At a frame rate of 30 frames per second, each window contains 45 frames and overlaps by two-thirds with adjacent windows. Window extraction was performed independently within each trajectory segment to avoid combining motion across discontinuities introduced by tracking gaps. The resulting windows serve as the basic units for local motion analysis.

For each window, a set of kinematic descriptors was computed from the smoothed palm landmark trajectories to capture multiple aspects of hand motion. Since all participants were right-hand dominant, descriptors were computed using the dominant hand only, ensuring consistency with the global motion representation and focusing on the primary hand involved in instrument manipulation. The descriptors capture key aspects of local motion, including translational behavior, trajectory efficiency, smoothness, dynamic intensity, and hand orientation dynamics [11, 49]. Motion smoothness is quantified using spectral arc length (SPARC), a frequency-domain measure shown to be more robust to noise than derivative-based metrics such as jerk [6]. An overview of the extracted features is provided in Table 4.2, with formal definitions given in Appendix 8.1.4.

In addition to the single-hand descriptors, an exploratory set of bimanual features was evaluated to capture inter-hand coordination patterns, including inter-hand distance statistics and velocity synchronization. However, these features require simultaneous tracking of both hands and reduced the number of usable motion windows due to missing detections. To maintain a consistent sample size, they were not included in the final modeling pipeline.

To prevent the local motion representation from being dominated by idle periods, windows with minimal motion were excluded. Such segments exhibit a low signal-to-noise ratio, where residual

Table 4.2: Local kinematic descriptors extracted for each 1.5s window.

Feature	Description	Motion aspect
Spatial spread	Standard deviation of palm position within the window	Workspace exploration
Path ratio	Ratio of traveled path length to start–end displacement	Trajectory efficiency
SPARC	Spectral arc length of velocity magnitude	Motion smoothness
Palm area CV	Coefficient of variation of palm triangle area	Hand orientation dynamics
HF ratio	Ratio of high-frequency to low-frequency spectral energy	Motion irregularity
Speed peaks	Number of prominent peaks in the speed profile	Sub-movement frequency
Acceleration RMS	Root mean square of acceleration magnitude	Dynamic intensity

tracking artifacts may outweigh meaningful motion, and are therefore unlikely to contribute reliably to skill assessment.

4.3.2 Statistical Aggregation of Local Features

Before introducing structured aggregation models, a simple statistical aggregation of window-based features was used as a baseline. Each video contains several hundred temporal windows; therefore, local kinematic descriptors were summarized at the video level using distribution statistics. For each feature, the 10th percentile, median, 90th percentile, and standard deviation were computed across all windows. For descriptors exhibiting heavy-tailed distributions, a $\log(1 + x)$ transformation was applied prior to aggregation to reduce the influence of extreme values.

The resulting aggregated descriptors were evaluated using the same protocol as for the global feature set. Spearman correlations with surgical skill scores were computed to assess univariate associations, followed by residual analysis based on partial R^2 to determine whether aggregated local descriptors provide information beyond the global baseline model.

This step serves as a diagnostic baseline, capturing the distribution of local motion characteristics without explicitly modeling temporal structure or recurring motion patterns. It provides a reference for assessing whether more structured approaches, such as Bag-of-Words representations or Multiple Instance Learning, benefit from their modeling assumptions.

The analysis further informed the construction of a compact feature set for subsequent local modeling. To reduce redundancy while retaining complementary information, intercorrelations between descriptors were examined. Based on this analysis, four descriptors were selected: spatial spread,

path ratio, palm area coefficient of variation, and SPARC smoothness, representing spatial extent, trajectory efficiency, orientation dynamics, and motion smoothness.

4.3.3 Motion Vocabulary Construction

To model recurring local motion patterns within surgical procedures, a Bag-of-Words (BoW) representation was constructed from the window-level kinematic descriptors introduced in the previous section. In this representation, individual motion windows are treated as observations in feature space and clustered into a discrete set of representative motion patterns, following established approaches to histogram-based motion representation [1, 14, 45]. Each video is subsequently represented by the relative frequency with which these primitives occur during the procedure.

Prior to clustering, all window-level descriptors were log-transformed using $\log(1 + |x|)$ to reduce the influence of heavy-tailed values. The transformed features were then standardized using robust scaling based on the median and interquartile range to account for remaining skewness in the feature distributions.

Motion primitives were discovered using k-means clustering applied to the normalized window feature vectors. This partitions the window-level observations into K clusters in feature space, with each cluster representing a recurring local motion pattern. In this context, the resulting clusters are interpreted as candidate motion primitives.

To obtain a stable video-level representation, a soft assignment scheme was used instead of assigning each window to a single cluster. Let d_{ik} denote the Euclidean distance between window i and cluster centroid k. Cluster membership probabilities were computed using a softmax transformation of the negative distances:

$$p_{ik} = \frac{\exp(-d_{ik})}{\sum_{j=1}^{K} \exp(-d_{ij})},$$

where p_{ik} represents the probability that window i belongs to cluster k.

For a video containing N motion windows, the Bag-of-Words representation is obtained by averaging these probabilities across all windows:

$$BoW_k = \frac{1}{N} \sum_{i=1}^{N} p_{ik}.$$

This results in a fixed-length feature vector of dimension K, where each component reflects the relative frequency of a discovered motion primitive within the procedure. From here on, the resulting Bag-of-Words dimensions are referred to as BoW features, each representing the prevalence of a specific motion primitive within the recorded procedure.

4.3.4 Hybrid BoW Model

To evaluate whether the discovered motion primitives contain predictive information beyond global motion efficiency, the Bag-of-Words representation was integrated into the global skill prediction model. Specifically, the BoW features derived from the movement vocabulary were combined with the previously established global predictors. The resulting hybrid model can be written as:

$$GRS \sim PC1 + CaseType + \sum_{k=1}^{K} BoW_k$$

In contrast to the earlier feature evaluation stages, which focused on assessing the incremental contribution of individual descriptors, the Bag-of-Words representation is intended to capture patterns in the joint distribution of local motion features. Meaningful predictive information may therefore arise from combinations of motion primitives rather than from individual local descriptors considered in isolation. For this reason, the BoW representation was evaluated using a model that considers all BoW features simultaneously. Model estimation was performed using LASSO-regularized regression, which imposes an L_1 penalty on the regression coefficients and thereby shrinks less informative predictors toward zero, yielding an implicit form of feature selection. The regularization strength was selected automatically by internal cross-validation within the evaluation protocol described in Section 4.2.3.

Since the optimal vocabulary size for representing motion primitives is not known a priori, the number of clusters K used for the Bag-of-Words construction was treated as a model parameter. Models were evaluated across a range of cluster sizes from $K = 8$ to $K = 25$ to assess how the granularity of the motion vocabulary influences predictive performance. This analysis was not intended to optimize a single hyperparameter value, but rather to evaluate whether the Bag-of-Words representation consistently improves skill prediction across different vocabulary sizes. The resulting performance trends therefore provide an estimate of the realistic improvement range that can be expected from incorporating motion primitives into the skill model.

In addition to the hybrid model, a second model using only the BoW features was evaluated to assess the extent to which motion primitives alone capture information about surgical skill.

4.3.5 Multiple Instance Learning with Gated Attention

While the Bag-of-Words representation captures recurring local motion patterns through a discrete vocabulary of motion primitives, it relies on a fixed clustering of feature space and aggregates information across segments uniformly. As a result, it does not explicitly model the varying importance of individual motion segments within a procedure. To address this limitation, a Multiple Instance Learning (MIL) framework with gated attention pooling was evaluated, which learns a continuous aggregation of window-level features without imposing a predefined discrete structure.

In the MIL formulation, a surgical video is treated as a bag of local motion windows, where each

window corresponds to a short temporal segment described by the four retained kinematic features (path ratio, spatial spread, SPARC, and palm area coefficient of variation), introduced in Section 4.3.1. The task of the model is to learn both how to aggregate these window-level features into a video-level representation and how to map this representation to the corresponding skill score.

Hybrid MIL Model To assess whether local motion representations provide information beyond global motion efficiency, the MIL framework was extended into a hybrid model combining aggregated window features with the global efficiency descriptor (PC1).

Each video is represented as a bag of local motion windows $\{x_i\}_{i=1}^{N}$. To obtain a fixed-length representation from this variable-length set, an attention-based pooling mechanism is applied. The model assigns a learnable importance weight α_i to each window and computes a weighted average representation:

$$z_{\text{local}} = \sum_{i=1}^{N} \alpha_i x_i, \quad \text{with} \sum_i \alpha_i = 1.$$

The attention weights are computed using the gated attention mechanism proposed by Ilse et al. [21], enabling input-dependent weighting of motion windows within each video.

The overall architecture is illustrated in Figure 4.2. Each video is processed along two parallel pathways. In the local pathway, the attention-pooled representation z_{local} is passed through a dedicated prediction head consisting of a linear layer and a nonlinearity producing a compact embedding of local motion structure. In parallel, the global feature vector, consisting primarily of the efficiency descriptor PC1, is processed by a separate branch of the same form.

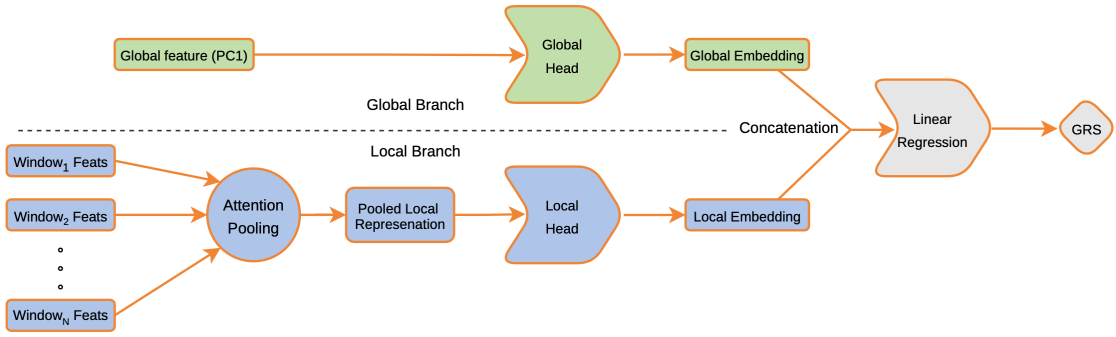

Figure 4.2: Overview of the hybrid Multiple Instance Learning (MIL) architecture. Local motion windows are aggregated using attention pooling to form a compact representation, which is processed through a local pathway. In parallel, the global efficiency descriptor (PC1) is processed through a separate global pathway. Both representations are concatenated and mapped to the final prediction using linear regression.

The use of separate pathways ensures that local and global signals are encoded independently before fusion, allowing their respective contributions to be assessed in a controlled manner. The outputs

of both branches are concatenated and mapped to the final prediction through a linear regression layer:

$$\hat{y} = f_{\text{fusion}}\big(f_{\text{local}}(z_{\text{local}}),\ f_{\text{global}}(g)\big),$$

where g denotes the global feature vector.

This design reflects the hypothesis that surgical skill is determined by global motion efficiency and the structure of local motion patterns, while maintaining an interpretable decomposition of both components.

Ablation Variants To isolate the contribution of different components of the model, several ablation variants were evaluated:

- **MIL-only model:** uses only the attention-pooled local windows without global features
- **Mean pooling model:** replaces attention with uniform averaging across windows
- **Hybrid mean pooling model:** combines mean-pooled local features with global features
- **Global-only model:** removes the local branch entirely

These variants allow the contribution of local motion information and the effect of learned attention-based aggregation to be assessed independently.

Training Procedure All models were trained using mean squared error loss and optimized with the Adam optimizer with weight decay. To prevent information leakage from the held-out test fold, all training decisions were based exclusively on training-set behavior. To reduce overfitting given the limited dataset size, a small model capacity and early stopping were employed. Training was stopped when either the training mean absolute error (MAE) reached a predefined threshold or no further improvement in training performance was observed for a fixed number of epochs. The MAE thresholds were selected based on preliminary experiments and informed by prior results from the global and Bag-of-Words models, which provided an estimate of the achievable performance range. Thresholds were defined per model category and kept fixed across comparable variants.

To further improve robustness, model parameters were stabilized using weight averaging over the final training epochs. In addition, for each cross-validation fold, five models were trained with different random initializations. The final prediction was obtained by averaging the outputs of these models, forming a small ensemble to reduce variance and improve robustness.

Summary This formulation enables the model to learn a data-driven aggregation of local motion patterns, allowing the relative importance of individual motion segments to be inferred directly from the data.

Chapter 5

Results

5.1 Tracking Reliability (RQ1)

To evaluate whether markerless hand tracking provides trajectories suitable for motion analysis, the coverage and temporal continuity of the extracted hand trajectories before and after post-processing were assessed. MediaPipe reliably detected hands in the majority of frames across the dataset. Raw detection coverage averaged $81.8 \pm 9.4\%$ for the left hand and $78.8 \pm 6.4\%$ for the right hand. However, while individual frame detections were generally accurate, occasional inconsistencies in handedness classification produced label switches between the left and right hands. In a trajectory context, even infrequent label swaps can result in large spatial discontinuities when detections are linked across time.

The trajectory post-processing pipeline described in Section 4.1 improved the temporal consistency of the tracked hand trajectories. After enforcing label consistency and removing implausible detections, small gaps in the trajectories were filled through interpolation. Following this procedure, temporally consistent trajectories could be reconstructed for $82.7 \pm 9.3\%$ of frames for the left hand and $79.1 \pm 6.4\%$ for the right hand. Only a small fraction of frames required interpolation ($1.31 \pm 0.89\%$ for the left hand and $0.66 \pm 0.40\%$ for the right hand), indicating that the vast majority of trajectory points correspond to directly observed hand positions.

Since the subsequent kinematic analysis relies on sliding windows of 1.5 seconds, the fraction of tracked frames belonging to trajectory segments of at least this duration was additionally evaluated. Across the dataset, $96.0 \pm 2.6\%$ of left-hand and $97.0 \pm 1.7\%$ of right-hand tracked frames were part of segments longer than 1.5 s, indicating that sufficiently long continuous trajectories were available for the window-based feature extraction.

Overall, these results suggest that markerless hand tracking provides sufficiently stable and temporally consistent motion trajectories for subsequent kinematic analysis when combined with the

proposed trajectory post-processing pipeline.

5.2 Global Motion Representation (RQ2)

5.2.1 Global Feature Screening

To obtain an initial assessment of the predictive potential of the extracted motion descriptors, Spearman correlations between each feature and the GRS skill scores were computed. The results are summarized in Figure 5.1. A subset of descriptors exhibits strong associations with surgical skill, with several features reaching absolute correlations above 0.5. The strongest signal is observed for right-hand angular direction switches ($|\rho| = 0.74$). In contrast, the remaining features show substantially weaker correlations ($|\rho| < 0.4$), indicating that the dominant skill-related signal is concentrated in a relatively small subset of descriptors.

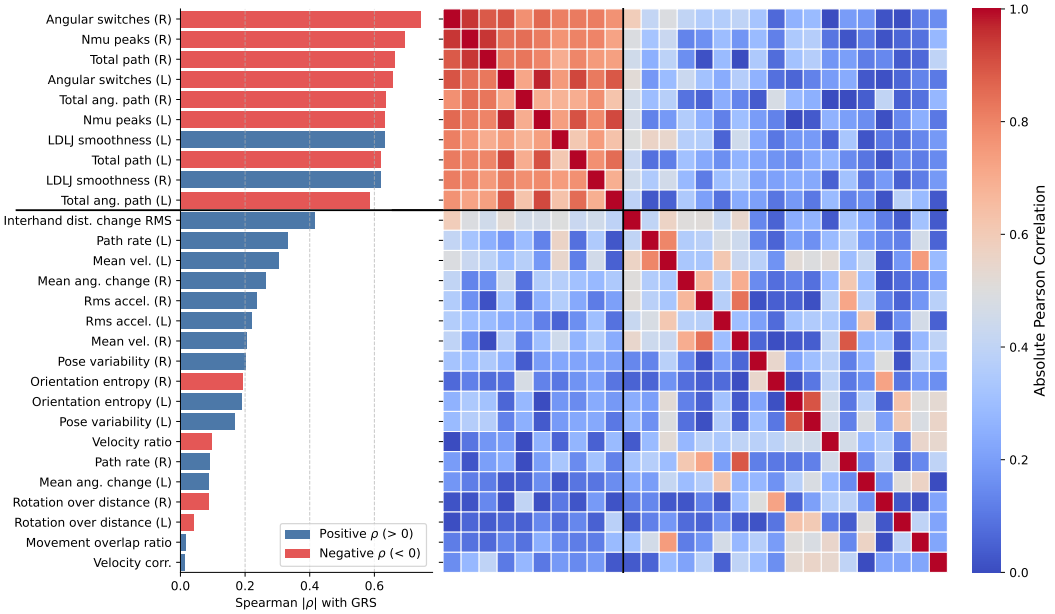

Figure 5.1: Feature screening results. Left: absolute Spearman correlation between each motion descriptor and the GRS skill score. Right: absolute intercorrelation matrix of the same descriptors. Highly predictive features form a tightly correlated block, indicating substantial redundancy.

The intercorrelation matrix in Figure 5.1 shows that the highly predictive descriptors form a tightly correlated cluster. Within this subset, pairwise correlations remain consistently high (minimum $\rho = 0.60$, mean 0.81 ± 0.10), indicating substantial redundancy. In contrast, the remaining descriptors exhibit weaker relationships both with GRS and with the high-signal cluster.

Feature–GRS correlations were computed on the full dataset for visualization. To verify robustness, the analysis was repeated within each LOSO training fold. Across all folds, the top descriptors consistently maintained correlations above $r > 0.5$, indicating that the observed signal is not driven

by individual surgeons.

Inspection of the highly correlated features shows that they primarily capture accumulated measures of motion activity, such as total path length, number of movement units, total angular displacement, and directional changes. These metrics reflect global motion efficiency, where more economical movement is associated with higher skill. A complementary descriptor is the log dimensionless jerk (LDLJ), which measures movement smoothness and exhibits a strong positive association with GRS, whereas the accumulation-based metrics show strong negative correlations.

Given the strong redundancy among these descriptors, Principal Component Analysis (PCA) was applied to obtain a compact representation of the shared signal. The first principal component (PC1) captures the dominant variation across these features and serves as a global motion efficiency index. Since all participants were right-hand dominant, two configurations were evaluated: one using only dominant-hand features and one combining both hands. The dominant-hand configuration yielded a higher explained variance (87.1% vs. 81.7%) and was therefore selected. PC1 is thus derived from right-hand descriptors capturing directional changes, movement segmentation, path length, angular displacement, and smoothness.

5.2.2 Baseline Model and Residual Analysis

Using PC1 as the sole predictor, a baseline skill model was constructed and evaluated using LOSO cross-validation (Section 4.2.3). The results are summarized in Table 5.1. Adding case type as a categorical predictor yields a modest improvement, likely reflecting differences in task duration and movement characteristics across procedures.

Table 5.1: Performance of global skill prediction models under leave-one-surgeon-out (LOSO) cross-validation.

Model	MAE	R^2	Spearman ρ
PC1 (Linear)	5.35	0.47	0.69
PC1 + Case Type (Ridge)	5.22	0.50	0.72
PC1 + Case Type (MLP)	4.96	0.52	0.71
Duration (Linear)	5.77	0.42	0.68
PC1 + Velocity Correlation (Ridge)	5.25	0.51	0.72

A small multilayer perceptron (MLP) was additionally evaluated using the same inputs. While a slight improvement in R^2 and MAE was observed, the gain was small and did not indicate strong nonlinear effects. Given its simplicity and interpretability, ridge regression was retained as the baseline model.

For comparison, a model using only task duration was also evaluated. Although duration captures an important aspect of efficiency, it yields lower predictive performance and reduced robustness

compared to motion-based descriptors. The PC1-based representation provides a more stable estimate by combining multiple correlated efficiency-related signals into a single predictor.

To assess whether additional descriptors provide information beyond global efficiency, residual feature analysis was performed using PC1 as the baseline predictor. Bimanual velocity correlation emerged as the strongest secondary feature, explaining approximately 10% of the remaining variance across LOSO training folds (partial $R^2 \approx 0.105$). However, its contribution decreased substantially when case type was included, indicating partial overlap with task-specific effects. A permutation-based residual test did not yield statistical significance ($p = 0.14$), suggesting that the observed improvement may be explained by random variation. Consequently, velocity correlation was not included in the conservative baseline model.

Nevertheless, the feature consistently received a non-zero regression weight when included alongside PC1 (mean coefficient $\beta \approx -2.12$), indicating that it captures residual variation conditional on global efficiency. The negative coefficient suggests that, among surgeons with similar efficiency levels, lower synchronization between hands is associated with higher skill. One possible explanation is that bimanual surgical tasks often involve differentiated roles between the hands, where one hand performs more active manipulation while the other provides support or stabilization. However, such role distributions are task-dependent and may vary with surgical context and level of expertise.

Overall, while bimanual coordination appears to contain additional signal, the effect remains exploratory given the limited dataset size and lack of statistical significance. The final baseline model therefore retains only the PC1-based efficiency representation and case type indicators.

5.3 Local Motion Modeling (RQ3)

5.3.1 Evaluation of Aggregated Local Features

Aggregated local features showed only moderate associations with GRS scores, with maximum absolute Spearman correlations of approximately $\rho \approx 0.45$. The strongest relationships were observed for descriptors related to trajectory variability and dynamic motion characteristics.

Residual analysis revealed that these features did not provide additional explanatory power beyond the global efficiency index (PC1), indicating that their predictive signal largely overlaps with the global representation. Overall, aggregated local descriptors contributed little beyond the baseline model (PC1 + case type). The median spatial spread yielded the highest residual contribution (mean partial $R^2 \approx 0.11$), but this effect was not statistically significant ($p = 0.30$). A LASSO regression model using the full set of aggregated descriptors (28 features) did not improve predictive performance and slightly reduced accuracy compared to the baseline, suggesting that high-dimensional aggregation introduces redundancy without additional signal.

Correlation analysis among descriptors confirmed substantial redundancy. High-frequency dynamics (HF Ratio and Speed Peaks) were strongly correlated ($r = 0.68$) and inversely related to motion

smoothness (SPARC), while measures of motion magnitude (Spatial Spread, Acceleration RMS, and Palm Area CV) also exhibited considerable overlap. Based on these observations, a compact set of four descriptors was retained for subsequent modeling: Spatial Spread, Path Ratio, Palm Area CV, and SPARC. These features capture complementary aspects of local motion, including spatial extent, trajectory structure, orientation dynamics, and smoothness.

Overall, the results indicate that simple statistical aggregation of local features provides limited additional predictive value, motivating the use of models that explicitly capture the structure of local motion patterns.

5.3.2 Hybrid BoW Model Performance

The predictive performance of the hybrid Bag-of-Words (BoW) model was evaluated across vocabulary sizes ranging from $K = 8$ to $K = 25$ (Figure 5.2). Across all tested values of K, the hybrid BoW model consistently outperformed the global baseline model ($PC1+$ case type), indicating that the motion primitive representation provides additional predictive signal. Performance varies moderately with K, with the best results observed at $K = 18$ ($R^2 \approx 0.58$).

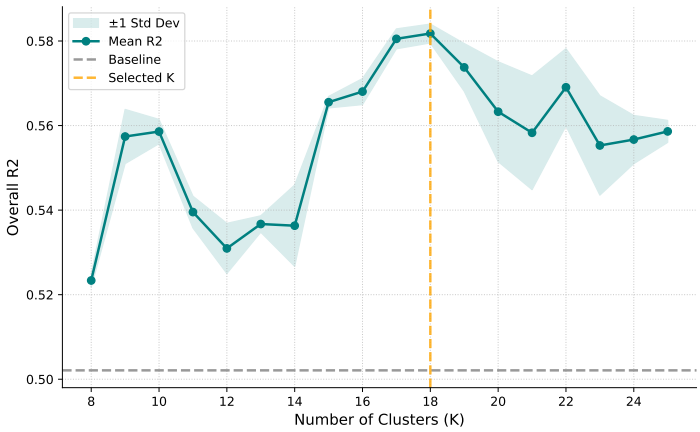

Figure 5.2: Cluster sensitivity analysis. Predictive performance of the hybrid Bag-of-Words model across different vocabulary sizes K. The shaded region indicates ± 1 standard deviation across clustering seeds. The dashed line marks the baseline $PC1+$ case type model.

Based on this analysis, $K = 18$ was selected as a representative vocabulary size. The corresponding model performance is summarized in Table 5.2. Since K was selected using the same cross-validation procedure, the reported results should be interpreted as optimistic estimates.

The hybrid BoW models substantially improve predictive performance compared to the global baseline. Including bimanual velocity correlation alongside PC1 leads to a further improvement, suggesting that coordination provides complementary information to both global efficiency and local motion patterns. However, this effect should be interpreted as exploratory again, as in the selection process of this feature no statistical significance was established.

Table 5.2: Performance of Bag-of-Words skill models under LOSO cross-validation. Hybrid models use $K = 18$, and BoW only model uses $K = 12$. Reported values are averaged over 10 runs with different k-means initializations.

Model	MAE	R^2	Spearman ρ
PC1 + Case Type + BoW	4.70	0.58	0.76
PC1 + Velocity Corr + BoW	4.65	0.62	0.80
BoW only	6.42	0.32	0.53

For comparison, models using only the BoW representation were also evaluated, over the same range of vocabulary sizes (Appendix 8.1). While these models achieve moderate performance ($R^2 \approx 0.32$), they remain substantially below the global baseline, indicating that motion primitives alone capture only part of the skill-related signal. However, the improvement observed in the hybrid models suggests that they provide complementary information when combined with global motion features. Interestingly, the optimal vocabulary size differs between the BoW-only and hybrid models, indicating that the motion primitives most predictive in isolation are not necessarily those that best complement global efficiency.

Overall, these results demonstrate that the BoW representation captures structured local motion patterns that enhance skill prediction when integrated with global motion descriptors.

5.3.3 Motion Primitive Interpretation

For the interpretation of the discovered motion primitives, a single, consistent set of cluster centroids is required. Under the LOSO protocol, clustering is performed independently within each fold, resulting in slight variations in centroid locations and permutations of cluster identities. To enable coherent interpretation, the clustering obtained from the full dataset is therefore used. To verify that this global clustering is representative of the fold-specific clusterings used during model evaluation, a stability analysis was performed. Cluster centroids obtained within each LOSO fold were aligned with the global clusters and their feature distributions were compared. Only minor variation was observed, indicating that the discovered motion primitives are stable across folds (Appendix 8.2).

Since the Bag-of-Words representation uses soft assignment, the BoW features reflect the average similarity of motion windows to each primitive rather than discrete occurrence counts. The LASSO model yields a sparse solution in which, in addition to the global efficiency index (PC1), only three motion primitives receive consistently non-zero coefficients across folds. All remaining primitives and the case-type indicators are effectively discarded, suggesting that they either capture skill-neutral motion patterns or redundant information. The exclusion of case-type variables further indicates that procedural differences are already reflected in the distribution of motion primitives.

The resulting model can be expressed as:

$$GRS \sim PC1 + BoW_4 + BoW_{11} + BoW_{13}.$$

Among the selected primitives, BoW_{13} exhibits the strongest effect ($\beta \approx -2.70$), followed by BoW_{11} ($\beta \approx -1.47$) and BoW_4 ($\beta \approx 0.86$). Notably, only BoW_{13} shows a substantial direct correlation with GRS ($\rho \approx -0.50$), whereas the other two primitives exhibit weak marginal associations. This indicates that their predictive contribution arises primarily in combination with other features, reflecting conditional relationships rather than direct effects.

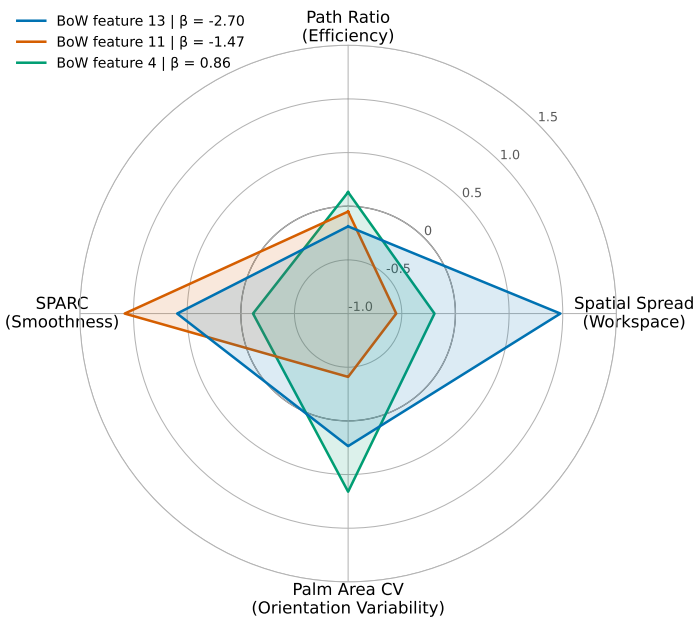

Figure 5.3: Kinematic profiles of the three motion primitives selected by the LASSO model. Feature values correspond to standardized cluster centroid values (log-transformed and robust-scaled). Positive values indicate above-average feature magnitudes relative to the dataset median. The SPARC feature was sign-flipped prior to the log-transform, such that smaller values correspond to smoother motion.

To better understand the motion characteristics captured by these primitives, Figure 5.3 visualizes their centroid feature profiles. The three selected primitives correspond to clearly distinct local motion regimes. BoW_{13} is characterized by large spatial spread and high SPARC values, indicating wide workspace exploration combined with irregular, non-smooth motion. This pattern is consistent with exploratory or corrective movements covering a large portion of the workspace. Given its strong negative association with GRS, this primitive can be interpreted as a novice-like movement pattern. In contrast, BoW_{11} exhibits low spatial spread and relatively stable hand pose, but high SPARC values indicating irregular motion. This suggests short, localized adjustments rather than large exploratory movements. BoW_4 occupies a different region of the feature space, combining moderate

trajectory efficiency with increased orientation variability. This points to motion segments involving noticeable pose adjustments within a spatially confined region. The absence of a clear expert–novice interpretation for BoW_{11} and BoW_4 is consistent with their weak marginal correlations and indicates that these primitives primarily contribute through conditional relationships with other motion characteristics.

5.3.4 Multiple Instance Learning Results

To evaluate whether learned aggregation of local motion windows provides additional predictive signal beyond global motion efficiency, the performance of MIL-based models was compared to mean pooling.

When using only local motion information, attention-based aggregation clearly outperforms mean pooling (Table 5.3), indicating that selectively weighting motion windows yields a more informative representation than uniform averaging. This suggests that not all motion segments contribute equally to skill prediction. However, this advantage disappears in the hybrid setting. When combined with the global efficiency descriptor (PC1), mean pooling performs comparably to, and even slightly outperforms, attention-based aggregation. This indicates that the information captured by attention overlaps substantially with the global efficiency signal, limiting its additional contribution.

Model	**MAE**	R^2	**Spearman** ρ
Mean pooling (local only)	6.97	0.21	0.43
Attention MIL (local only)	6.24	0.35	0.56
Mean pooling + PC1	4.45	0.60	0.76
Attention MIL + PC1	4.64	0.58	0.75

Table 5.3: Performance comparison of MIL-based aggregation and mean pooling, both in isolation and in combination with PC1.

Analysis of attention weights provides further insight into this behavior. In the MIL-only setting, attention weights are highly selective, with large disparities between the most and least important windows (on the order of 10^2). In contrast, attention becomes more uniform in the hybrid setting, indicating reduced reliance on identifying specific informative segments when global features are available.

To investigate whether this behavior arises because the strong global predictor limits the ability of the local branch to learn complementary representations, or instead reflects redundancy with the global efficiency descriptor, two additional strategies were evaluated. First, a two-stage training procedure was considered, in which the MIL model was initially trained using only local motion windows and subsequently fine-tuned after introducing the global feature branch. Second, a residual learning approach was evaluated, where the MIL model was trained to predict the residuals of the baseline

ridge model, and the final prediction was obtained by combining both outputs. Neither approach improved performance over the standard hybrid model, suggesting that the limited contribution of attention reflects a lack of complementary information beyond PC1 rather than an imbalance in the learning process.

Including case type in the global branch did not improve performance and slightly reduced accuracy, consistent with earlier findings that task-related differences are already captured by motion representations.

Overall, these results show that while local motion patterns contain meaningful structure, their contribution is largely redundant with global motion efficiency when both are considered jointly.

5.3.5 Diagnostic Hybrid Model Analysis

Motivated by the observation that mean pooling performs comparably to attention-based aggregation in the hybrid setting, an additional post-hoc diagnostic experiment was conducted to assess whether predictive performance is primarily driven by the underlying aggregated local features or by the modeling architecture itself. To this end, local motion information was represented using the median window values of the four selected descriptors (Spatial Spread, Path Ratio, Palm Area CV, and SPARC), which were combined with the global efficiency descriptor PC1 in a ridge regression model:

$$\text{GRS} \sim \text{PC1} + \text{mdn(Spatial Spread)} + \text{mdn(Path Ratio)} + \text{mdn(Palm Area CV)} + \text{mdn(SPARC)}$$

This model achieves performance comparable to the best-performing aggregation-based models ($MAE = 4.72$, $R^2 = 0.60$, $\rho = 0.78$). This is consistent with the ablation results observed in the MIL setting, where hybrid models converge to similar performance levels regardless of the aggregation strategy. Notably, earlier analyses based on larger statistically aggregated feature sets did not reveal this effect. In contrast, the present result indicates that comparable performance can be obtained by combining a small number of carefully selected local descriptors with PC1, without relying on more complex aggregation mechanisms or modeling approaches. For completeness, a model using only the selected median-aggregated local descriptors was also evaluated. It achieves moderate predictive performance ($MAE = 6.30$, $R^2 = 0.37$, $\rho = 0.60$), indicating that these features capture meaningful skill-related information in the absence of the global efficiency component.

While aggregate performance metrics indicate that different modeling approaches converge to similar accuracy, they do not reveal how predictions are distributed across the range of skill scores. To better characterize model behavior, predicted versus true GRS scores are shown for representative models in Figure 5.4. Panel (a) corresponds to a linear model based solely on PC1, while Panel (b) shows the diagnostic model combining PC1 with median-aggregated local motion descriptors.

In the baseline model (Panel a), predictions exhibit a clear regression-to-the-mean effect, with higher scores slightly underestimated and lower scores slightly overestimated. This tendency is

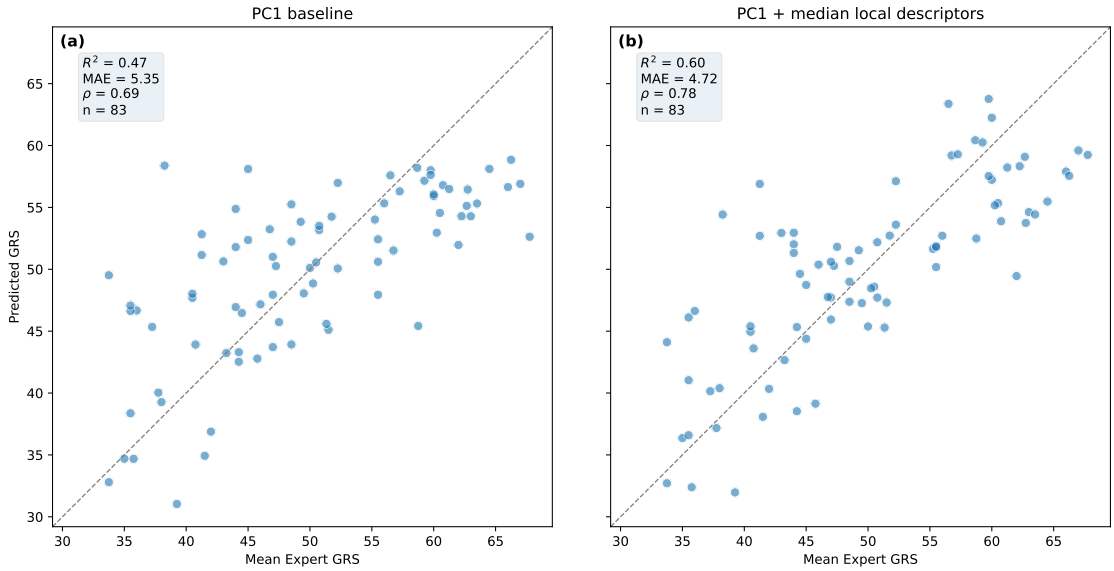

Figure 5.4: Predicted versus true mean expert GRS scores for (a) the PC1-based linear baseline and (b) a ridge model combining PC1 with median-aggregated local motion descriptors. The dashed line indicates perfect agreement.

reduced in the hybrid model (Panel b), where predictions align more closely with the identity line.

The figure provides a visual confirmation of the modest but consistent improvement obtained by incorporating aggregated local motion descriptors, while also illustrating that the dominant structure of the predictions is already captured by the global efficiency signal.

Chapter 6

Discussion

6.1 Interpretation of Results

The results of this study provide a consistent picture of how surgical skill in orthopedic trauma training is reflected in motion patterns derived from computer vision–based hand tracking. In particular, they address three key aspects: the feasibility of extracting reliable motion signals, the extent to which global motion representations explain skill, and the additional contribution of local motion structure.

Markerless hand tracking was found to provide motion data of sufficient stability and quality for kinematic analysis when combined with appropriate preprocessing. While raw trajectories are affected by noise, detection errors, and temporal inconsistencies, the applied processing pipeline yields smooth and physically plausible motion signals. This supports the use of vision-based hand tracking as a viable source of motion data in controlled training environments, despite its inherent limitations.

At the representational level, the results indicate that surgical skill is largely captured by a low-dimensional global motion component. A substantial portion of the variance in expert-rated performance can be explained by a single latent dimension derived from correlated kinematic descriptors. This dominant component reflects global motion efficiency, consistent with prior observations that expert performance is characterized by more economical and controlled movement [4, 9, 11]. These findings suggest that skill-related variation is highly structured and can be summarized compactly at the level of whole-procedure motion.

Local motion patterns, in contrast, provide a more nuanced but comparatively secondary contribution. Initial analyses based on Bag-of-Words representations suggested that only a subset of motion primitives consistently contributes to prediction performance, indicating that certain types of movement may be more strongly associated with surgical skill. However, this interpretation is

not fully supported by subsequent experiments. Evidence from Multiple Instance Learning and the diagnostic model shows that performance gains from incorporating local information are largely independent of the specific aggregation strategy. This suggests that skill-relevant information is not concentrated in a small number of highly informative segments, but is instead distributed across the procedure. Consequently, it is the aggregated form of local motion descriptors—capturing patterns across the entire procedure—that provides the most useful signal. In this form, they encode meaningful skill-related information on their own, while most importantly complementing the dominant global efficiency signal in ways not captured by the initial set of global descriptors. A plausible explanation for this behavior is that global descriptors forming PC1 summarize accumulated motion characteristics, but cannot distinguish whether variations arise from true differences in movement efficiency or from confounding factors such as tracking variability or task-specific differences. In this context, aggregated local descriptors may provide this additional contextual information, contributing a modest but consistent improvement in predictive performance.

The interpretation of model performance should be considered in the context of variability in expert ratings. The aggregated GRS score exhibits a standard error of measurement (SEM) of approximately 2.9 points, reflecting inherent uncertainty in the target variable. As a result, part of the prediction error is attributable to label noise rather than model limitations. A mean absolute error of approximately 4.7 therefore corresponds to a deviation exceeding expected rating variability by less than 2 points. Notably, model predictions are slightly more consistent than individual expert ratings, whose average disagreement is approximately 4.8 points.

Taken together, these findings indicate that surgical skill, as captured through hand motion in this setting, is primarily expressed as a global property of movement that manifests throughout the procedure. while local motion patterns provide additional structure, their contribution is primarily complementary and results in a significant and consistent improvement in predictive performance. Across the explored approaches, similar performance gains are observed once a small set of informative window-level descriptors is incorporated, largely independent of the specific aggregation or modeling strategy, suggesting that predictive performance is driven more by the quality of the underlying motion descriptors than by model complexity.

6.2 Limitations and Methodological Considerations

Limitations of Representation Learning An exploratory attempt was made to learn local motion representations using an unsupervised autoencoder framework. The motivation was to reduce reliance on handcrafted feature design and to investigate whether compact representations of short motion segments could be learned directly from the data. A lightweight 1D convolutional autoencoder with a low-dimensional bottleneck was trained to reconstruct invariant kinematic signals extracted from 1.5-second sliding windows of hand trajectories, yielding a large corpus of over 50,000 unlabeled motion segments.

However, the learned latent representations did not provide meaningful skill-related signal in subsequent analyses. Inspection of the reconstruction outputs revealed significant signal attenuation: while input trajectories exhibited dynamic and multi-modal behavior, the reconstructed signals were consistently oversmoothed with reduced dynamic range. This indicates that the model primarily captures dominant motion patterns while suppressing finer-scale variations.

This limitation appears to arise from a combination of factors. First, the reconstruction objective encourages preservation of overall signal structure rather than subtle, discriminative variations that may be relevant for skill assessment. Second, the low-dimensional bottleneck imposes a strong compression constraint, limiting the ability to retain nuanced kinematic characteristics. Finally, the aggregation of temporal features into a compact latent representation reduces sensitivity to the temporal organization of motion within each window.

In contrast, the handcrafted kinematic descriptors used in this work explicitly encode motion characteristics known to be relevant for surgical skill, such as smoothness, variability, and coordination. The results therefore suggest that, in this setting, domain-informed feature engineering combined with structured aggregation provides a more reliable representation of local motion than unsupervised reconstruction-based embeddings. More generally, these findings highlight the importance of aligning representation learning objectives with the downstream task, for example through supervised or contrastive approaches, rather than relying on reconstruction alone.

Limitations of Vision-Based Motion Tracking An additional limitation of this work arises from the use of markerless, vision-based hand tracking. While this approach enables non-intrusive data collection, it is inherently less precise than sensor-based or optical tracking systems. The extracted trajectories are subject to noise, occlusions, and frame-to-frame estimation variability, which necessitates post-processing steps such as smoothing and interpolation to obtain stable motion signals [30].

As a consequence, fine-grained kinematic characteristics, such as tremor, micro-adjustments, or subtle variations in movement smoothness, may be attenuated or lost in the processed trajectories. This has direct implications for the type of skill-related information that can be captured. In particular, the resulting representations are favorable to global motion properties and overall movement structure, while limiting the ability to model high-frequency or fine-scale aspects of motor control.

Researcher Bias in Small Data Settings One of the key methodological insights of this work was the importance of research integrity in a setting where research quality is largely assessed based on predictive performance. While the goal was never to intentionally report inflated results, the process of model development revealed how easily performance can be inadvertently optimized through iterative design choices when working with a fixed dataset.

In this study, the same dataset serves as the basis for both model development and evaluation under a cross-validation protocol. Although strict separation between training and test data is

maintained within each fold, repeated exposure to the dataset during model development introduces a more subtle form of overfitting at the level of the research process. Design decisions such as data transformations, feature selection, and model configuration can become implicitly influenced by observed performance across multiple iterations. Early in the project, this form of "manual overfitting" was not explicitly considered, under the assumption that preventing direct data leakage between training and test sets would be sufficient to ensure valid evaluation.

However, this assumption proved incomplete. Any design decision informed by observations on the full dataset, or adjusted primarily to improve performance without clear justification grounded in domain knowledge or data structure, risks introducing bias into the reported results. A concrete example arises in the construction of the global efficiency index (PC1). An initial approach selected features based on their minimum absolute correlation with GRS scores computed across all LOSO folds. While this resulted in slightly improved performance, it is not methodologically sound, as it implicitly incorporates information from the test portion of each fold into the feature selection process. A more principled approach is to restrict such decisions to the training data within each fold. However, even under this constraint, additional challenges arise. For example, the choice of a correlation threshold for feature inclusion remains somewhat arbitrary and can itself be tuned to maximize performance.

To address this, the methodology in this work was revised to ensure that all design decisions were consistent with the evaluation protocol. In the case of PC1, feature selection was not originally driven by the goal of maximizing performance, but by identifying descriptors with consistently strong associations to GRS scores. While correlations reported in this work are computed on the full dataset for descriptive purposes, the selection of the feature subset itself does not depend critically on these full-data observations. The same set of top-performing descriptors is identified within the training data of each LOSO fold, indicating that the selection is stable with respect to the specific data partitioning. This stability suggests that the selected feature subset reflects a robust signal rather than an artifact of the particular dataset used in this study. In particular, the magnitude of the correlations and the clear separation between the strongest features and the remaining descriptors indicate that the same subset would likely be identified under new but comparable data conditions. Furthermore, the selected descriptors are not arbitrary, but capture closely related aspects of motion efficiency, providing conceptual coherence to the resulting representation.

These observations highlight a broader limitation of small-data research: even when rigorous evaluation protocols such as leave-one-subject-out cross-validation are used, the risk of implicit bias remains when the same dataset is repeatedly used to guide model development. Methodological rigor in this setting therefore extends beyond the choice of evaluation protocol. Ensuring the validity of results requires not only preventing data leakage, but also carefully constraining the entire model development process through transparency, statistical validation, and justification-driven decision making, so that reported performance reflects genuine predictive signal.

6.3 Future Directions

A primary limitation of motion-based skill assessment is that it captures how a task is performed, but not whether it is performed correctly. While the proposed approach extracts meaningful skill-related signal from motion, it does not directly assess outcome quality, such as fracture alignment or screw placement. Future work should therefore explore the integration of outcome-based measures to complement motion-derived representations. In controlled training settings, this could be achieved through image-based comparison, through quantivative alignment, between the post-procedure state and a known reference configuration. Combining such outcome-based signals with motion representations would allow for a more comprehensive assessment of both procedural quality and task success.

A second direction concerns the deployment of the proposed approach in real-world or training environments. While the current pipeline demonstrates that markerless hand tracking can be performed efficiently, the models rely on data collected under controlled conditions with a fixed camera setup. Future work should therefore focus on improving robustness to variations in viewpoint, camera placement, and environmental conditions. This should involve incorporating additional spatial information, such as depth estimation, as well as training on more diverse datasets covering multiple recording configurations. Such improvements are essential for enabling reliable deployment in practical training or clinical settings.

A third direction concerns learned representations. The reconstruction-based autoencoder explored in this work did not yield useful representations for surgical skill assessment, suggesting a mismatch between the learning objective and the downstream task. Future work could investigate alternative objectives that are better aligned with skill-related variation, such as contrastive, or weakly supervised approaches. In particular, models that explicitly capture temporal structure or incorporate may enable the learning of more informative motion representations [2, 27, 42].

Closely related to this is the development of more interpretable and actionable feedback for surgical training. While global performance scores provide a quantitative assessment of skill, they do not directly indicate which aspects of a procedure require improvement. A promising direction is to combine motion analysis with action recognition, where procedures are segmented into distinct movement types such as drilling, screwing, or repositioning [14, 22].

Segment-level motion representations could then be used to characterize how different types of actions are performed. Rather than requiring explicit segment-level annotations, weakly supervised or multiple-instance learning approaches could relate these local motion patterns to overall skill. This would enable more fine-grained and actionable feedback by identifying which categories of movements are associated with higher or lower levels of performance. In the longer term, such representations may also support the prediction of structured evaluation criteria, such as individual OSATS dimensions, thereby linking quantitative motion analysis to clinically interpretable assessment.

Chapter 7

Conclusion

This thesis investigated the use of markerless hand tracking as a basis for motion-based surgical skill assessment in orthopedic trauma training. The central objective was to determine whether vision-based hand motion data can provide meaningful and reliable information about surgical skill, how this information is structured, and how it can be effectively represented for prediction.

First, the results show that markerless hand tracking can provide motion data of sufficient quality for kinematic analysis when combined with appropriate preprocessing. Despite the inherent noise and variability of vision-based tracking, the extracted trajectories form a stable and physically plausible representation of hand motion, enabling the computation of meaningful motion descriptors. This demonstrates the feasibility of using standard video data as a scalable alternative to specialized sensing systems in this context.

Second, the analysis of global motion representations reveals that a large portion of skill-related variation can be captured by a single latent dimension derived from correlated kinematic features. This dominant component, interpreted as global motion efficiency, provides a compact and interpretable representation of surgical performance and explains a substantial fraction of the variance in expert ratings.

Third, the investigation of local motion structure shows that window-level motion descriptors contain additional, complementary information about skill. However, this information is distributed across the procedure rather than concentrated in a small number of highly informative segments. Across different modeling approaches, including Bag-of-Words, Multiple Instance Learning, and simple aggregated feature models, similar performance levels are achieved once global efficiency is combined with a small set of informative local descriptors. This indicates that predictive performance is driven primarily by the choice and quality of motion descriptors, while the specific aggregation or modeling strategy plays a comparatively minor role.

Taken together, these results provide a coherent picture of how surgical skill is reflected in hand

motion: skill is primarily expressed as a global property of movement that manifests consistently throughout the procedure, while local motion patterns provide complementary information that refines this representation. This work advances the state of the art by demonstrating the feasibility of fully vision-based, hand-centric motion analysis for orthopedic surgical skill assessment and by clarifying the relationship between global efficiency and local motion structure in this setting.

At the same time, several limitations remain. The proposed approach captures how a task is performed, but does not directly assess outcome quality, such as the correctness of fracture alignment or implant placement. In addition, the models are developed and evaluated under controlled recording conditions, and their robustness to real-world variability remains to be established. Furthermore, while local motion features provide complementary information, their interpretation remains indirect, limiting the ability to generate actionable feedback for trainees.

Future work should therefore focus on integrating outcome-based measures, improving robustness to diverse recording conditions, and developing methods for more interpretable and fine-grained feedback. In particular, combining motion analysis with action recognition and structured evaluation criteria may enable more informative assessment systems that not only quantify performance but also guide skill acquisition.

Overall, this work shows that markerless hand tracking provides a viable and scalable foundation for motion-based surgical skill assessment, and demonstrates that skill is primarily encoded in global efficiency-related motion characteristics, while aggregated local motion representations provide complementary information that leads to improved predictive performance.

Code Availability

The code used in this work is available at: GitHub repository.

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Chapter 8

Appendix

8.1 Mathematical Definition of Motion Descriptors

This appendix provides the mathematical definition of the motion descriptors used in the global and local analyses. All descriptors are computed from smoothed 2D hand landmark trajectories extracted from video. Unless otherwise stated, the dominant hand trajectory is represented by the palm center position

$$\mathbf{x}_t = (x_t, y_t), \qquad t = 1, \ldots, T,$$

sampled at frame rate f_s (here $f_s = 30$ Hz), with temporal step $\Delta t = 1/f_s$.

Frame-to-frame displacement is defined as

$$d_t = \|\mathbf{x}_t - \mathbf{x}_{t-1}\|_2,$$

Velocity, acceleration, and jerk magnitudes are derived from the displacement signal as

$$v_t = \frac{d_t}{\Delta t_t}, \qquad a_t = \frac{v_t - v_{t-1}}{\Delta t_t}, \qquad j_t = \frac{a_t - a_{t-1}}{\Delta t_t},$$

where Δt_t denotes the elapsed time between frames, accounting for possible tracking gaps in the global representation.

8.1.1 Palm orientation and pose representation

For descriptors related to hand orientation, a rigid palm representation is defined using landmarks

$$\mathbf{p}_t^{(0)} \text{ (wrist)}, \qquad \mathbf{p}_t^{(5)} \text{ (index MCP)}, \qquad \mathbf{p}_t^{(17)} \text{ (pinky MCP)}.$$

Two palm vectors are constructed as

$$\mathbf{v}_t^{(1)} = \mathbf{p}_t^{(5)} - \mathbf{p}_t^{(0)}, \qquad \mathbf{v}_t^{(2)} = \mathbf{p}_t^{(17)} - \mathbf{p}_t^{(0)}.$$

The in-plane palm orientation angle is defined as

$$\theta_t = \operatorname{atan2}\big(v_{t,y}^{(1)}, v_{t,x}^{(1)}\big).$$

A pose-related scalar proportional to the signed palm triangle area is defined through the 2D cross product

$$c_t = v_{t,x}^{(1)} v_{t,y}^{(2)} - v_{t,y}^{(1)} v_{t,x}^{(2)}.$$

For the local features, the palm area is computed as

$$A_t = \frac{1}{2}|c_t|.$$

8.1.2 Global motion descriptors

Total duration For each hand, the tracked duration is

$$D = \sum_{t=1}^{T} \Delta t_t.$$

Total path length The total traveled path length is

$$L = \sum_{t=1}^{T} d_t.$$

Mean velocity The mean translational velocity is

$$\bar{v} = \frac{1}{T}\sum_{t=1}^{T} v_t.$$

RMS acceleration Dynamic intensity is quantified by the root-mean-square acceleration:

$$a_{\mathrm{RMS}} = \sqrt{\frac{1}{T}\sum_{t=1}^{T} a_t^2}.$$

Path rate The path rate normalizes traveled distance by total duration:

$$\text{PathRate} = \frac{L}{D}.$$

LDLJ smoothness Movement smoothness is quantified using the log dimensionless jerk (LDLJ), implemented as

$$\text{LDLJ} = -\log\left(\frac{D^3}{L^2}\sum_{t=1}^{T} j_t^2\,\Delta t + \varepsilon\right),$$

where $\varepsilon = 10^{-6}$ is a small constant introduced for numerical stability.

Number of movement units (NMU) Movement segmentation is approximated by counting prominent peaks in the velocity profile. Let $\mathscr{P}$ denote the set of peaks detected in $\{v_t\}$ using a prominence threshold τ_v. Then

$$\text{NMU} = |\mathscr{P}|.$$

In the implementation, $\tau_v = 30$ pixels/s.

Mean absolute angle change Let

$$\phi_t = \text{atan2}(y_t - y_{t-1},\, x_t - x_{t-1})$$

be the direction of frame-to-frame displacement. Directional change is measured through wrapped angular differences

$$\Delta\phi_t = \text{wrap}(\phi_t - \phi_{t-1}),$$

where $\text{wrap}(\cdot)$ maps angles to $(-\pi, \pi]$. The mean absolute angle change is

$$\text{MAAC} = \frac{1}{|\mathscr{T}|}\sum_{t\in\mathscr{T}} |\Delta\phi_t|,$$

where $\mathscr{T}$ contains time points with sufficient movement amplitude.

Angular direction switches Angular switches quantify the number of sign reversals in the wrapped turning-angle sequence $\Delta\phi_t$, after excluding low-motion samples. This feature captures repeated alternation in local trajectory curvature.

Total angular path The accumulated palm rotation is defined as the total wrapped change in orientation angle:

$$\text{TAP} = \sum_{t=2}^{T} \left|\text{wrap}(\theta_t - \theta_{t-1})\right|.$$

Orientation entropy Palm orientation entropy is computed from a histogram of θ_t over B bins spanning $(-\pi, \pi]$. If p_b denotes the normalized histogram value in bin b, then

$$H_\theta = -\sum_{b=1}^{B} p_b \log p_b,$$

where zero-probability bins are omitted from the sum. In the implementation, $B = 36$.

Pose variability Pose variability is quantified as the coefficient of variation of the palm cross-product magnitude:

$$\text{PoseVar} = \frac{\text{std}(c_t)}{\text{mean}(|c_t|)}.$$

Rotation over distance This descriptor normalizes rotational motion by translational motion:

$$\text{RotDist} = \frac{\text{TAP}}{L + \varepsilon},$$

with $\varepsilon = 10^{-6}$.

8.1.3 Global bimanual coordination descriptors

For bimanual features, left and right hand trajectories are aligned by frame index. Let $v_t^{(L)}$ and $v_t^{(R)}$ denote left and right hand velocities, and let

$$\mathbf{x}_t^{(L)} = (x_t^{(L)}, y_t^{(L)}), \qquad \mathbf{x}_t^{(R)} = (x_t^{(R)}, y_t^{(R)})$$

denote the corresponding palm center positions.

Velocity correlation Temporal synchronization between hands is quantified using the Pearson correlation coefficient:

$$\rho_v = \text{corr}\left(v_t^{(L)}, v_t^{(R)}\right).$$

Movement overlap ratio The overlap ratio measures the fraction of aligned frames in which both hands are moving above threshold:

$$\text{Overlap} = \frac{1}{T} \sum_{t=1}^{T} \mathbf{1}\left(v_t^{(L)} > \tau_v \wedge v_t^{(R)} > \tau_v\right),$$

where $\tau_v = 30$ pixels/s.

Inter-hand distance change RMS The inter-hand distance is

$$r_t = \left\| \mathbf{x}_t^{(L)} - \mathbf{x}_t^{(R)} \right\|_2.$$

Its temporal stability is quantified as

$$\mathrm{RMS}_{\Delta r} = \sqrt{\frac{1}{T-1}\sum_{t=2}^{T}(r_t - r_{t-1})^2}.$$

Velocity ratio Relative hand activity is summarized by

$$\mathrm{VelRatio} = \frac{\bar{v}^{(L)}}{\bar{v}^{(R)} + \varepsilon},$$

with $\varepsilon = 10^{-6}$.

8.1.4 Local window-based descriptors

For the local analysis, each trajectory segment is partitioned into overlapping windows of fixed duration. Within each window, features are computed from the dominant hand only.

Let a window contain N samples. Let $\mathbf{x}_t = (x_t, y_t)$ denote the wrist landmark trajectory within the window, and define velocity and acceleration vectors as

$$\mathbf{v}_t = \frac{d\mathbf{x}_t}{dt}, \qquad \mathbf{a}_t = \frac{d\mathbf{v}_t}{dt},$$

implemented numerically using finite differences via `numpy.gradient`. Their magnitudes are

$$\|\mathbf{v}_t\|_2 = \sqrt{v_{x,t}^2 + v_{y,t}^2}, \qquad \|\mathbf{a}_t\|_2 = \sqrt{a_{x,t}^2 + a_{y,t}^2}.$$

Path ratio Trajectory efficiency within a window is defined as the ratio between traveled path length and net displacement:

$$\mathrm{PathRatio} = \frac{\sum_{t=1}^{N} d_t}{\|\mathbf{x}_N - \mathbf{x}_1\|_2 + 1}.$$

A constant of 1 pixel is added in the denominator for numerical stability.

Spatial spread Workspace exploration is quantified as the combined standard deviation of the 2D position:

$$\mathrm{SpatialSpread} = \sqrt{\mathrm{std}(x_t)^2 + \mathrm{std}(y_t)^2}.$$

SPARC Motion smoothness is quantified using spectral arc length (SPARC), computed from the magnitude spectrum of the speed signal

$$s_t = \|\mathbf{v}_t\|_2.$$

Let $P(f)$ denote the periodogram-based power spectral density estimate of s_t, and let

$$A(f) = \frac{\sqrt{P(f)}}{\max_f \sqrt{P(f)}}$$

denote the normalized amplitude spectrum. Restricting to frequencies $f \le f_c$, the spectral arc length is computed as

$$\text{SPARC} = -\sum_{k=1}^{K-1} \sqrt{(\Delta f_k)^2 + (\Delta A_k)^2},$$

where $f_c = 10$ Hz in the implementation. Less negative SPARC values correspond to smoother motion.

High-frequency ratio Motion irregularity is quantified from the tapered speed signal using the ratio of high-frequency to total band-limited spectral energy. Let $P_s(f)$ denote the power spectral density of the Tukey-windowed speed signal. Then

$$E_{\text{low}} = \sum_{f \in [0.5, 3.0]} P_s(f), \qquad E_{\text{high}} = \sum_{f \in (3.0, 10.0]} P_s(f),$$

and

$$\text{HF Ratio} = \frac{E_{\text{high}}}{E_{\text{low}} + E_{\text{high}} + \varepsilon},$$

with $\varepsilon = 10^{-9}$.

Speed peaks Sub-movement frequency is approximated by the number of prominent peaks in the speed profile:

$$\text{SpeedPeaks} = |\mathscr{P}_s|,$$

where $\mathscr{P}_s$ is the set of peaks detected in s_t using a prominence threshold proportional to the within-window speed variability.

Acceleration RMS Dynamic intensity is quantified by

$$\text{AccRMS} = \sqrt{\frac{1}{N} \sum_{t=1}^{N} \|\mathbf{a}_t\|_2^2}.$$

Palm area coefficient of variation Local pose variability is measured from the palm triangle area A_t as

$$\text{PalmAreaCV} = \frac{\text{std}(A_t)}{\text{mean}(A_t)},$$

with the convention that the feature is set to zero when $\text{mean}(A_t)$ is numerically negligible.

8.1.5 Implementation note

The mathematical definitions above reflect the implemented feature extraction pipeline. Some descriptors, especially those involving peak detection, spectral estimation, or wrapped angles, depend not only on the mathematical form of the feature but also on implementation choices such as smoothing, numerical differentiation, thresholding, frequency cutoffs, and histogram binning. These implementation details are therefore part of the operational definition of the features used in this work. All features were implemented as described below; code is available at [repo link].

8.2 Supplementary Analyses

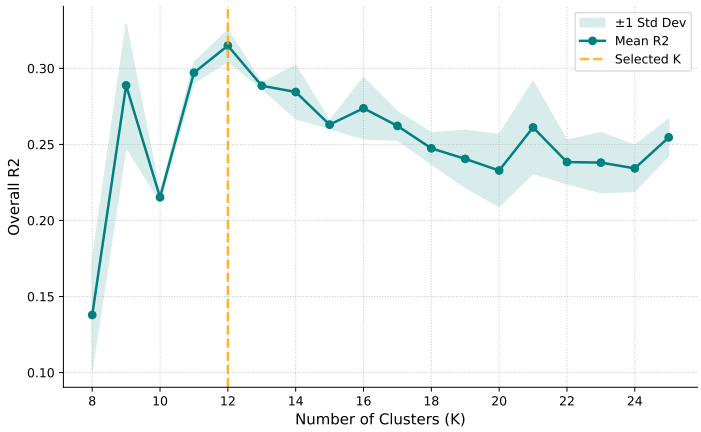

Figure 8.1: Cluster sensitivity analysis for the Bag-of-Words model using only BoW features. Predictive performance is shown across vocabulary sizes K. The shaded region indicates ± 1 standard deviation across clustering seeds.

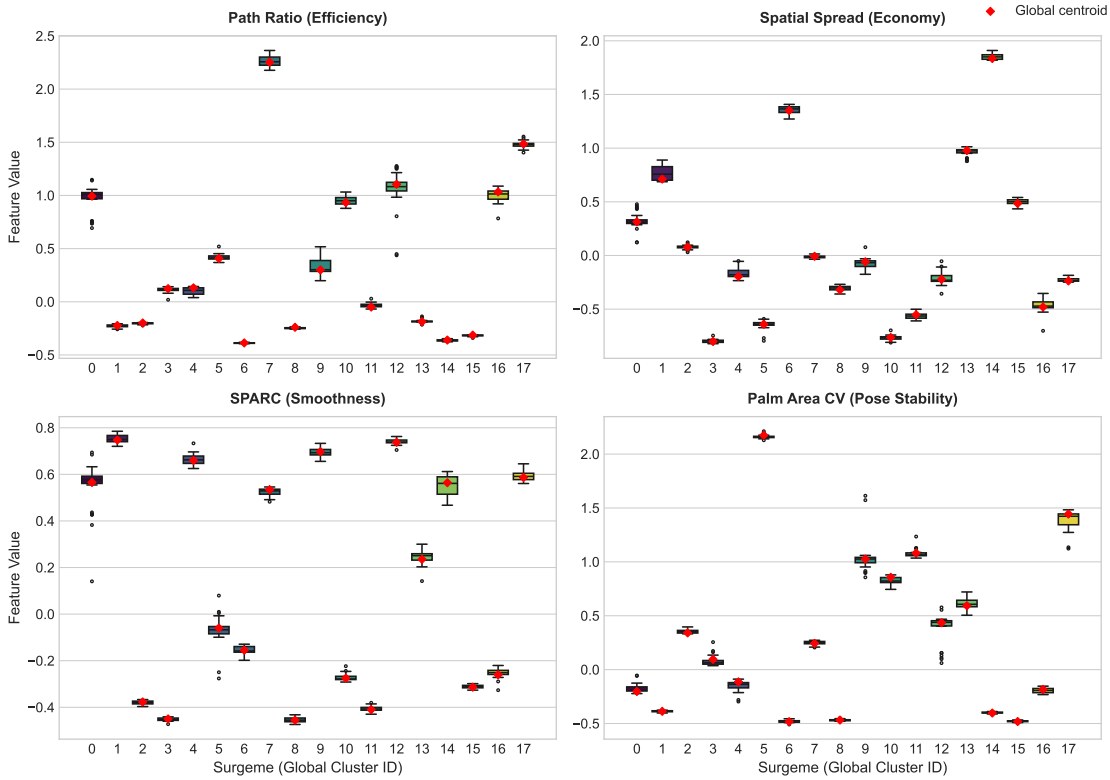

Figure 8.2: Stability of Bag-of-Words cluster centroids across LOSO folds. Centroids obtained from individual training folds are compared to those derived from the full dataset. The observed consistency indicates that the learned motion primitives are relatively stable with respect to variations in the training data.